

ATTACHMENT C – GROUNDWATER MONITORING AND LIMITATIONS

Permit Number: WQ0004332

Version: 4.1

Monitoring Wells: MW-1, MW-2, MW-3, MW-4, MW-6, and MW-7

GROUNDWATER CHARACTERISTICS		GROUNDWATER STANDARDS		MONITORING REQUIREMENTS		
PCS Code	Parameter Description	Daily Maximum		Frequency Measurement	Sample Type	Footnotes
00680	Carbon, Tot Organic (TOC)		mg/L	3 x Year	Grab	1, 6
00940	Chloride (as Cl)	250	mg/L	3 x Year	Grab	1
31616	Coliform, Fecal MF, M-FC Broth, 44.5 °C		#/100 mL	3 x Year	Grab	1
00610	Nitrogen, Ammonia Total (as N)	1.5	mg/L	3 x Year	Grab	1
00620	Nitrogen, Nitrate Total (as N)	10	mg/L	3 x Year	Grab	1
00400	pH	6.5-8.5	su	3 x Year	Grab	1, 2
00665	Phosphorus, Total (as P)		mg/L	3 x Year	Grab	1
70300	Solids, Total Dissolved - 180 °C	500	mg/L	3 x Year	Grab	1
GWVOC	Volatile Compounds (GW)		Present: Yes/No	Annually	Grab	1, 4, 5
82546	Water Level, Distance from measuring point		feet	3 x Year	Calculated	1, 2, 3

- 3 x Year monitoring shall be conducted in April, August, and November; Annual monitoring shall be conducted in November.
- The measurement of water levels shall be made prior to purging the wells. The depth to water in each well shall be measured from the surveyed point on the top of the casing. The measurement of pH shall be made after purging and prior to sampling for the remaining parameters.
- The measuring points (top of well casing) of all monitoring wells shall be surveyed to provide the relative elevation of the measuring point for each monitoring well. The measuring points (top of casing) of all monitoring wells shall be surveyed relative to a common datum.
- Volatile Organic Compounds (VOC) - In November only, analyze by one of the following methods:
 - Standard Method 6230D, PQL at 0.5 µg/L or less
 - Standard Method 6210D, PQL at 0.5 µg/L or less
 - EPA Method 8021, Low Concentration, PQL at 0.5 µg/L or less
 - EPA Method 8260, Low Concentration, PQL at 0.5 µg/L or less
 - Another method with prior approval by the Water Quality Permitting Section Chief

Any method used shall meet the following qualifications:

 - A laboratory shall be DWR certified to run any method used.
 - The method used shall include all the constituents listed in Table VIII of Standard Method 6230D.
 - The method used shall provide a PQL of 0.5 µg/L or less that shall be supported by laboratory proficiency studies as required by the DWR Laboratory Certification Unit. Any constituents detected above the MDL but below the PQL of 0.5 µg/L shall be qualified (estimated) and reported.
- If any volatile organic compounds (VOC) are detected as a result of monitoring as provided in Attachment C, then the Washington Regional Office supervisor, telephone number (252) 946-6481, shall be contacted immediately for further instructions regarding any additional follow-up analyses required.
- If TOC concentrations greater than 10 mg/L are detected in any downgradient monitoring well, additional sampling and analysis shall be conducted to identify the individual constituents comprising this TOC concentration. If the TOC concentration as measured in the background monitor well exceeds 10 mg/L, this concentration will be taken to represent the naturally occurring TOC concentration. Any exceedances of this naturally occurring TOC concentration in the downgradient wells shall be subject to the additional sampling and analysis as described above.
- Monitoring wells shall be reported consistent with the nomenclature and location information provided in Figure 3 and this attachment.